Code Comprehension

What Is Code Comprehension?

- (what questions do YOU think are important to answer?)

Importance of Reading

- "Two recent studies found that developers spend, on average, 58% and 70% of their time trying to comprehend code but only 5% of their time editing it."

- Unpublished experiment with Copilot: Copilot reduced task completion time by 18.5%

- 18.5 > 5; explain?

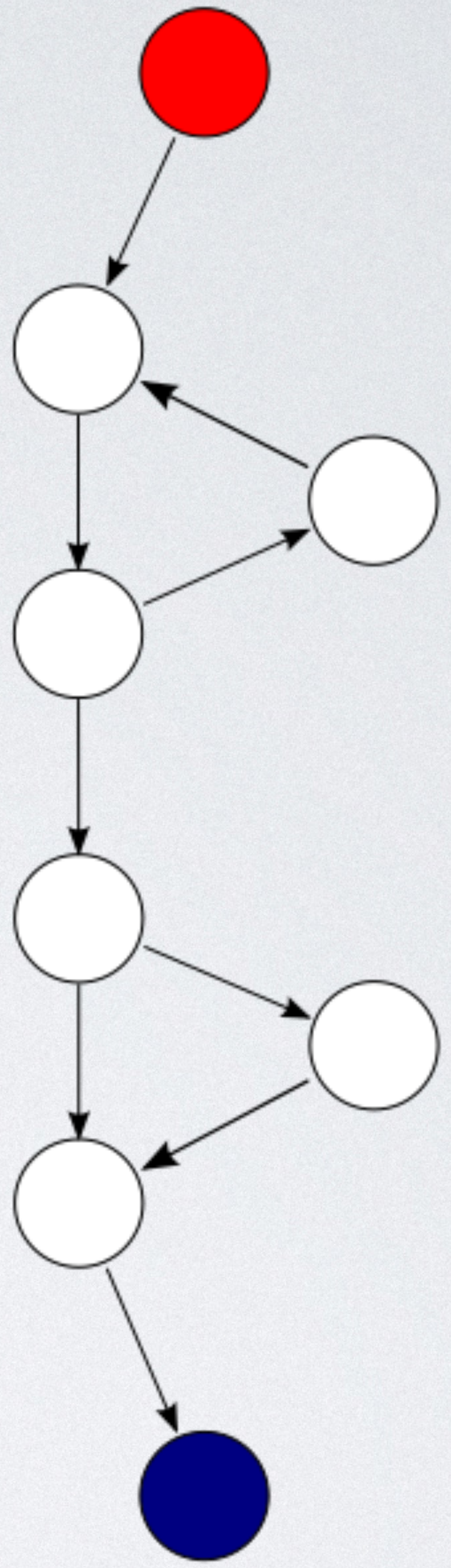
    -

McCabe Cyclomatic Complexity (1976)

Considering the control flow graph:

M = E - N + 2P

- E = the number of edges of the graph.
- N = the number of nodes of the graph.
- P = the number of connected components.



$9 - 8 + 2 \times 1 = 3$

Source: https://en.wikipedia.org/wiki/Cyclomatic_complexity

But What Does It all Mean?

- "Syntax, predicates, idioms — what really affects code complexity?"
  - "`for` loops are significantly harder than `if` s, that some but not all negations make a predicate harder, and that loops counting down are slightly harder than loops counting up."

Studies

- Controlled Experiments on Loops
- Eye Tracking to Study Code Regularity
- The Effect of Variable Names
- Also consider McCabe's cyclomatic complexity

For Each Study:

- In groups, summarize:
  - Research question
  - Method (tasks? participant population?)
  - Results
- Analyze:
  - Internal Validity
  - External Validity

Agenda

- Is there hope of a resulting theory?
- What should be done next? More experiments? If not, what instead?